# Introduction to Search Algorithm and Graph Algorithm

Week 3

# Search : ① The process of navigating a problem space to find a solution.

② AI agents use search algorithms to decide which actions to take in an environment.

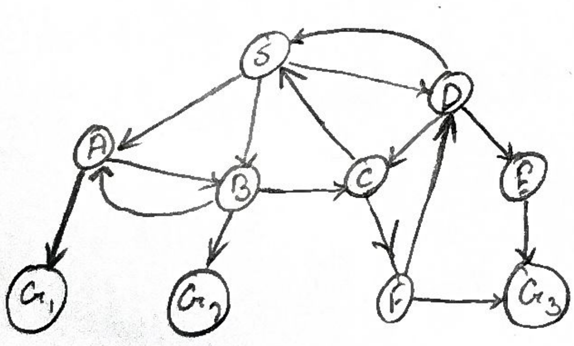
→ Types of search :

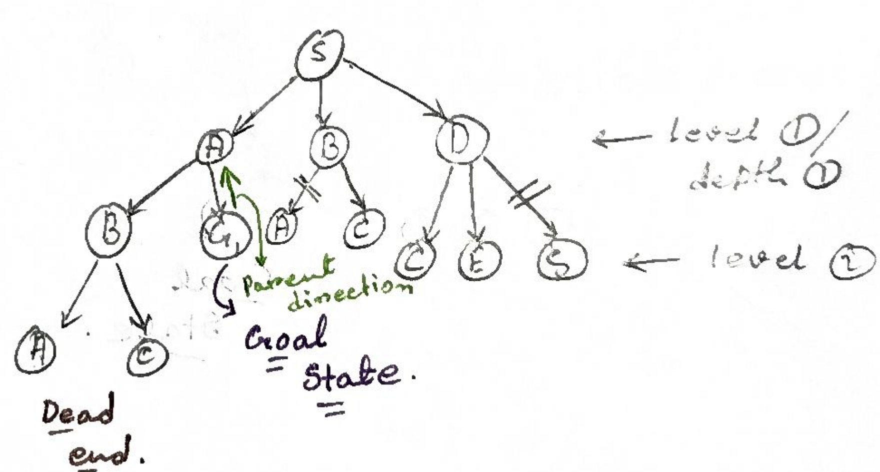
① Uninformed (Blind) Search

② Informed (Heuristic) search.

# Breadth - First Search :

- Explores all nodes at one depth before moving deeper
- Uses a queue (FIFO) to manage nodes.
- Generatees the shortest path (if all actions have the same cost).




Visited : S, A, B, D

path : S → A → $G_1$

shortest path.

# BFS stops when reach Goal state.

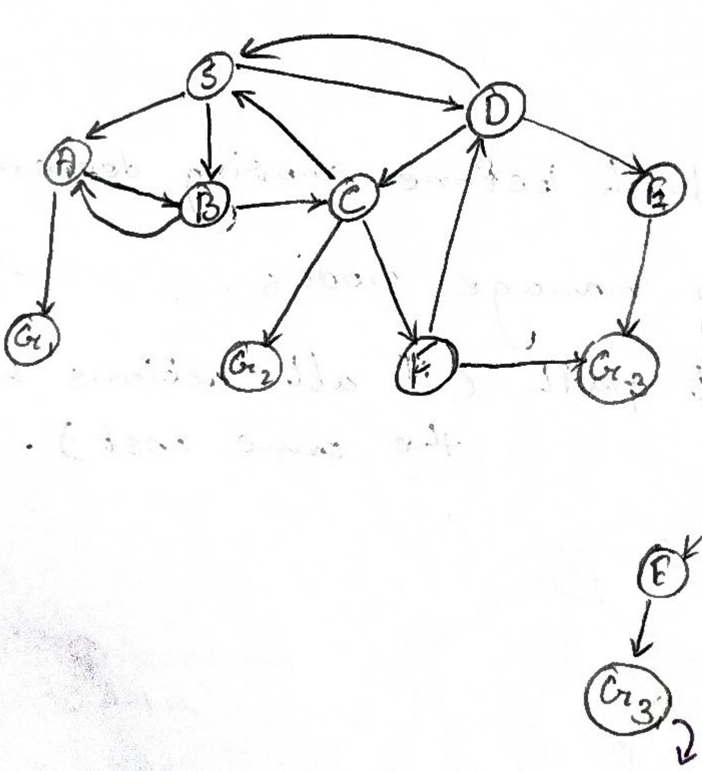
# Depth - First Search :
uninformed se

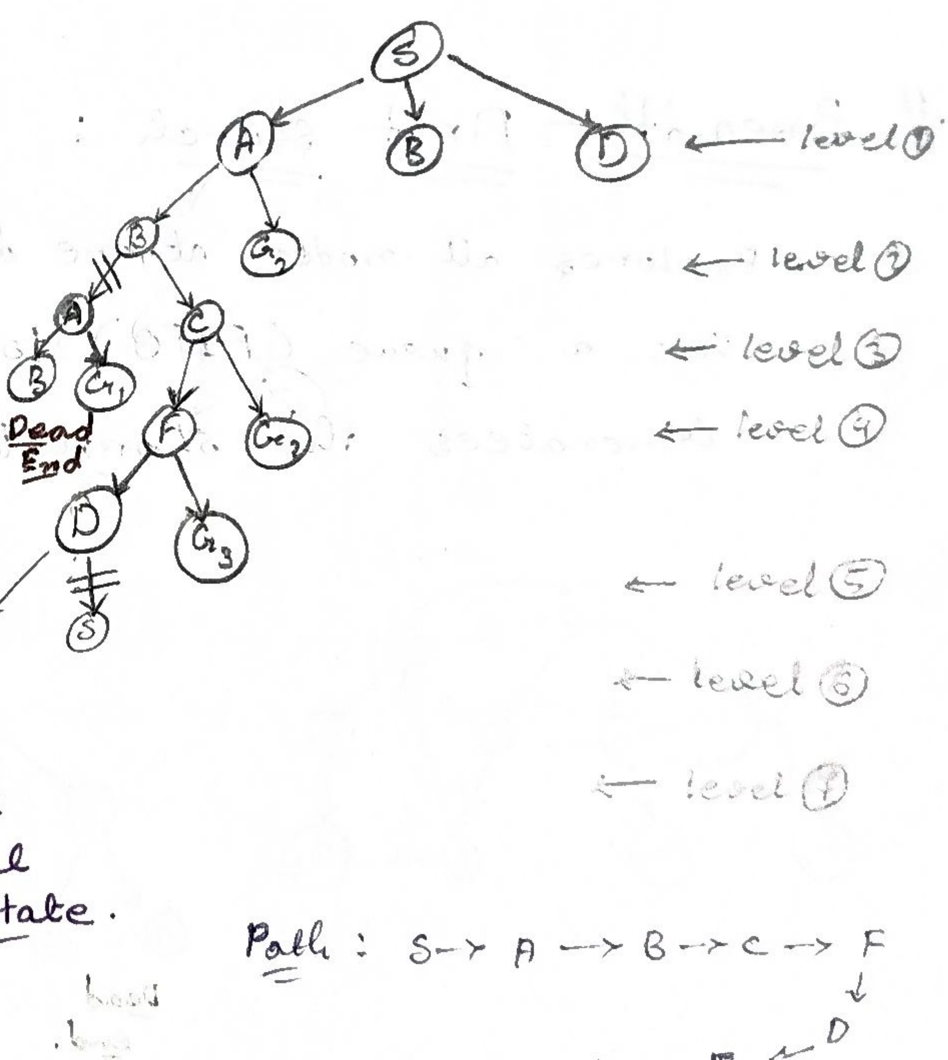
① Explore as deep as possible along a branch before backtracking.

② Use a stack (LIFO) for node management (can be recursive).

③ Memory - efficiency but may not find the shortest path.

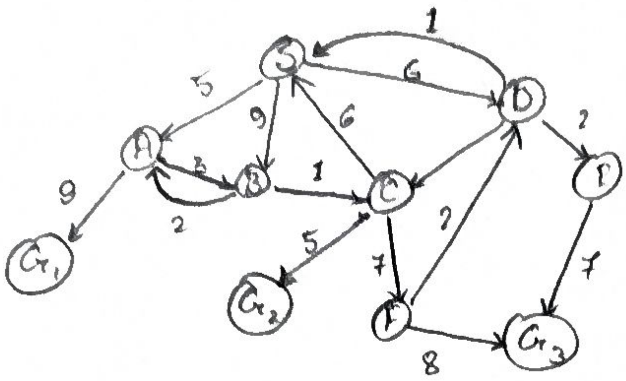
Time Complexity : $O(b^d)$
Space Complexity : $O(bd)$

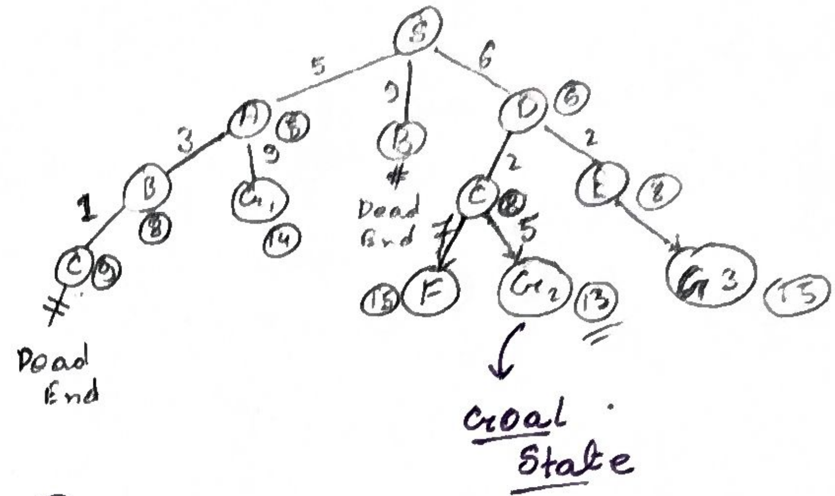



Path : S → A → B → C → F → D → E → $G_3$

# Search Space and Weighted Graph :

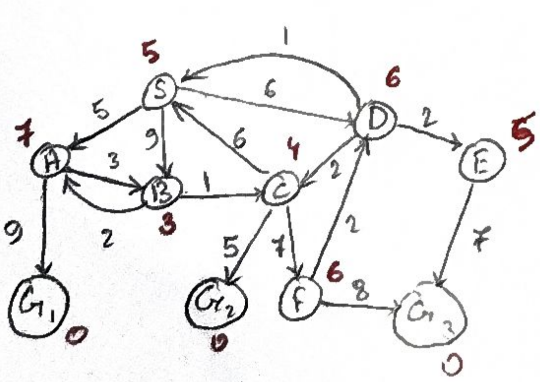


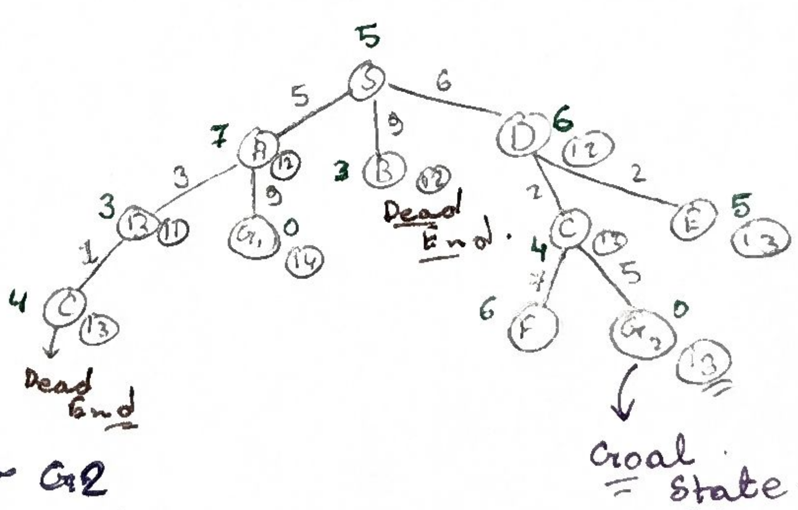



Path → S → D → C → G2

# A* Search Algorithm :
Informed Search

- Combines UCS with heuristics : $f(n) = g(n) + h(n)$.
- $g(n)$ : Cost to reach node.
  $h(n)$ : estimated cost to goal (heuristic)
- Finds optimal path if heuristic is admissible (never overestimates).






Path : S → D → C → G2